

SpectroElectroChem Suite

User Manual - Version 2.0

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This manual describes installation, program structure, input formats and the main analysis workflows.

Acknowledgement

Parts of the source code were developed with the assistance of OpenAI ChatGPT and were subsequently validated, modified and extended by the author.

1. Overview

SpectroElectroChem Suite is a software package for spectro-electrochemical data analysis. It provides modules for Raman spectra, SERS/Raman voltammograms, absorptovoltammograms and fluorovoltammograms.

The central application contains a menu bar, toolbar and module cards. Each module can be started independently from the main window.

2. Modules

Module	Purpose	Main outputs
Raman Spectrum Analysis	Baseline correction, smoothing, peak detection	Excel, PNG, PDF, HTML
SERS / Raman Voltammogram	3D and 2D visualization of potential-dependent Raman data	Excel, HTML
Absorpto- / Fluorovoltammogram	Analysis of wavelength- and potential-dependent absorption or fluorescence data	Excel, HTML

3. Installation

For Python users, install the required packages with:

```
pip install -r requirements.txt
```

Then start the program with:

```
python main.py
```

On Windows, the file Start_SpectroElectroChem_Suite.bat can be used.

4. Building a Windows application

The package includes a PyInstaller specification file and a build script. On Windows, run:

```
scripts\build_exe_windows.bat
```

The generated application is placed in dist\SpectroElectroChem_Suite.

5. Input formats

Raman spectrum analysis expects a CSV file where the first column contains Raman shifts and the remaining columns contain intensity spectra.

Voltammogram modules use matrix CSV files. The first row contains potentials, the first column contains Raman shifts or wavelengths, and the remaining cells contain measured intensities, absorption values or fluorescence values.

6. Data processing

The Raman module supports ModPoly, AsLS, arPLS and airPLS baseline correction methods. Savitzky-Golay smoothing is available for reducing spectral noise while preserving peak shapes.

The voltammogram modules generate interactive 3D surfaces, heatmaps, contour plots and rotatable waterfall plots. Excel files include matrix data, long-format data and, where applicable, waterfall values for custom plotting.

7. Plugin architecture

The plugin registry is stored in src/spectroelectrochem_suite/plugins/plugins.json. Future modules, for example IR or UV/Vis spectro-electrochemistry, can be added by creating a Python module with a main() function and registering it in this JSON file.

8. GitHub and releases

The repository includes a GitHub Actions workflow for building a Windows application. After uploading to GitHub, replace the placeholder YOUR_GITHUB_NAME in the updater and citation files. Creating a tag such as v2.0.0 can trigger an automatic build.

9. Citation

If you use the software in teaching, reports or publications, please cite the repository or DOI and mention the software version.

Habekost, A. SpectroElectroChem Suite, Version 2.0.0.